9.5 Probability of Disjoint and Overlapping Events

Name _____ Period ____ Date _____

Algebra I 9.5 · Student sheet

I can…

- tell whether two events are disjoint or overlapping.
- find P(A or B) when A and B are disjoint.
- find P(A or B) when A and B overlap.

Vocabulary · fill in the missing word

Compound event — two events _____ by "and" or "or"

Disjoint (mutually exclusive) events — events that can't _____ at the same time

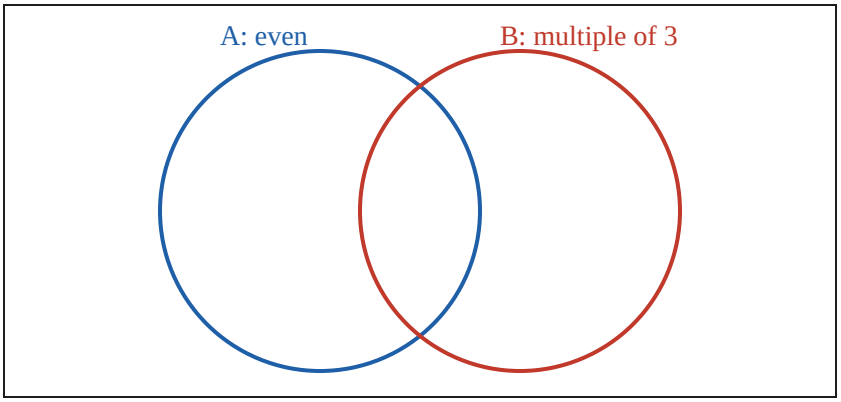
Overlapping events — events that have at least one _____ in common

Disjoint — P(A or B) = _____

Overlapping — P(A or B) = _____

Part 1 · Venn sort

Write each number from 1 to 20 in the correct region. Numbers in neither circle go outside the circles.



Count in A	Count in B	Count in both	Count in A or B

P(A) + P(B) = _____ P(A or B) = _____ Which numbers did you count twice? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Disjoint events can't _____. P(A or B) = _____

Example 1

A card is chosen from cards numbered 1 to 10. What is the probability that the number is less than 3 or greater than 8?

Example 2

The spinner has 8 equal sections. What is the probability of spinning red or an even number?



Overlapping events: P(A or B) = _____

Why subtract?: I subtract P(A and B) so the overlap is counted only _____.

Example 3

In a class of 30 students, 18 play a sport, 12 are in the band, and 5 do both. What is the probability that a student plays a sport or is in the band?

"Or" and "and": For "or" I _____; for "and" I _____.

Example 4

At a theater, 60% of the audience are adults. 30% of the adults and 50% of the children say they would see the movie again. What is the probability that a randomly chosen audience member would see it again?

Part 3 · Partner practice

Partner 1

You roll a die. What is the probability of rolling an even number or a number greater than 4?

Partner 2

A bag has 4 red, 5 blue, and 3 green marbles. What is the probability of choosing a red or a green marble?